

Math 150 – Exam 3 – Chapter 4 – Fall 2015

I pledge to demonstrate personal and academic integrity in all matters with my fellow students, the faculty, and the administration here at USD. I promise to be honest and accountable for my actions; and to uphold the stature of scholastic honesty to better myself and those around me. Initials: _____

Name: _____ Section: **1 2**

Instructions: Read all directions carefully. Ask for clarification if you do not understand the directions. Show your work when necessary. Complete sentences and justifications are only necessary when indicated. There are 6 pages to this exam.

There are 100 points on this test and 4 Extra Credit points.

Num.	Score	Out of
1	3	3
2		15
3		12
4		20
5		20
6		20
7		10
B		*4*
Total		100

1. {3 points} Take a deep breath. Smile for 4 seconds.

2. {3 points each} State whether each statement is True or False.

- (a) Let $a < b$. Suppose f is continuous on the closed interval $[a, b]$, differentiable on the open interval (a, b) , and $f(a) = f(b)$. Then there is a c in (a, b) with $f'(c) = 0$.

TRUE

- (b) If $f'(x) = 0$ when $x = 3$, then $f(x)$ is either a local minimum or a local maximum.

False

- (c) If $f(a)$ is an absolute maximum and $f'(a) = 0$, then $f(a)$ is a local maximum.

True

- (d) If $f'(x) = g'(x)$ for all real numbers x , then $f(x) = g(x)$.

False

- (e) If $\lim_{x \rightarrow a} f(x) \cdot g(x)$ is of the indeterminate form $\infty \cdot 0$, then

False

$$\lim_{x \rightarrow a} f(x) \cdot g(x) = \lim_{x \rightarrow a} f'(x) \cdot g'(x).$$

3. {12 points} Fill in the blanks:

- (a) If $f'(x) > 0$ for $-2 < x < 5$, then f is increasing on the interval $(-2, 5)$.

- (b) If $f''(2) < 0$ and $f'(2) = 0$, then f is concave down when $x = 2$ and $f(2)$ is a local max of f .

- (c) If $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$, then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ is in indeterminate form $\frac{0}{0}$

and by L'Hopital's rule, $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$.

3	/	15	/	12
3				

4. {4 points each} Evaluate the limits. Use l'Hopital's rule if necessary and possible. Be sure to show all your work.

(a) $\lim_{x \rightarrow 2} \frac{x^2 - 3x + 2}{x - 2}$ IFOT $\frac{0}{0}$ or $\lim_{x \rightarrow 2} \frac{(x-2)(x-1)}{(x-2)} = \lim_{x \rightarrow 2} x-1 = 1$

$\lim_{x \rightarrow 2} \frac{2x-3}{1} = 4-3 = 1$

(b) $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$ by definition. or

IFOT $\frac{\infty}{\infty}$

$\ln y = \ln \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = \lim_{n \rightarrow \infty} n \ln \left(1 + \frac{1}{n}\right) = \lim_{n \rightarrow \infty} \frac{\ln \left(1 + \frac{1}{n}\right)}{\frac{1}{n}}$ IFOT $\frac{0}{0}$

$= \lim_{n \rightarrow \infty} \frac{1}{1 + \frac{1}{n}} = 1$ so $\ln y = 1$ so $y = e$

(c) $\lim_{x \rightarrow \pi/2^-} \sec x - \tan x$

$\lim_{x \rightarrow \pi/2^-} \frac{1}{\cos x} - \frac{\sin x}{\cos x} = \lim_{x \rightarrow \pi/2^-} \frac{1 - \sin x}{\cos x}$ IFOT $\frac{0}{0}$

$\lim_{x \rightarrow \pi/2^-} \frac{-\cos x}{-\sin x} = \frac{-0}{-1} = 0$

(d) $\lim_{x \rightarrow 0^+} x \ln x$ $0 \cdot (-\infty)$

$\lim_{x \rightarrow 0^+} \frac{\ln x}{1/x}$ IFOT $\frac{\infty}{\infty}$ $\lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} = \lim_{x \rightarrow 0^+} -x = 0$

(e) $\lim_{x \rightarrow \infty} x e^{-x^2/2} = \lim_{x \rightarrow \infty} \frac{x}{e^{x^2/2}}$ IFOT $\frac{\infty}{\infty}$

$\lim_{x \rightarrow \infty} \frac{1}{e^{x^2/2} (x)} = \frac{1}{\infty} = 0$

5. Consider the function $f(x) = xe^{-x^2/2}$. The first derivative of f is $f'(x) = (1 - x^2)e^{-x^2/2}$ and the second derivative of f is $f''(x) = x(x - \sqrt{3})(x + \sqrt{3})e^{-x^2/2}$.

[Additional Hints]: $\sqrt{3} \approx 1.7$, and $f(\sqrt{3}) \approx 0.4$, $\lim_{x \rightarrow -\infty} xe^{-x^2/2} = 0$ and f is an odd function.

- (a) {2 points} What are the critical values of f .

$$x = 1, -1$$

- (b) {2 points} List the intervals in which f is increasing.

$$(-1, 1)$$

- (c) {2 points} List the intervals in which f is decreasing.

$$(-\infty, -1) \cup (1, \infty)$$

- (d) {2 points} List any local minimums and local maximums.

$$\begin{aligned} \max f(1) &= e^{-1/2} \\ \min f(-1) &= -e^{-1/2} \end{aligned}$$

- (e) {2 points} List the intervals in which f is concave up.

$$(-\sqrt{3}, 0) \cup (\sqrt{3}, \infty)$$

- (f) {2 points} List the intervals in which f is concave down.

$$(-\infty, -\sqrt{3}) \cup (0, \sqrt{3})$$

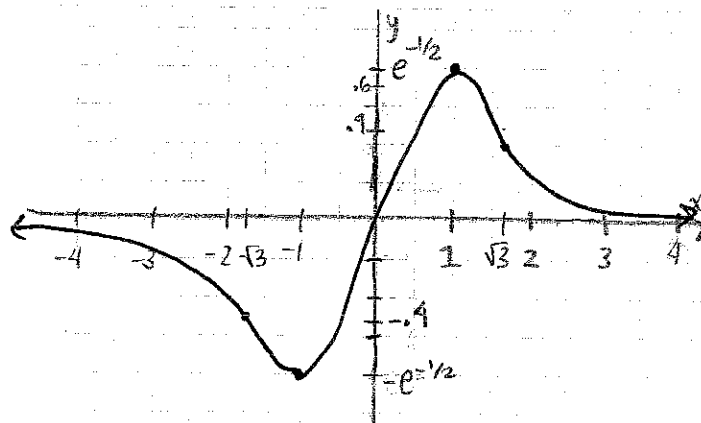
- (g) {2 points} Give any inflection points of f .

$$(-\sqrt{3}, f(\sqrt{3})), (0, f(0)), (\sqrt{3}, f(\sqrt{3}))$$

- (h) {2 points} State any vertical or horizontal asymptotes of f .

$$y = 0 \text{ is a HA}$$

- (i) {4 points} Graph the function f on the graph below.



6. {4 points each} Find the most general antiderivatives.

(a) $f(x) = 5x^{1/4} - 7x^{3/4}$

$$F(x) = 4x^{5/4} - 4x^{7/4} + C$$

(b) $f(x) = \frac{10}{x^9} = 10x^{-9}$

$$F(x) = -\frac{10}{8}x^{-8} = -\frac{5}{4}x^{-8} = -\frac{5}{4x^8} + C$$

(c) $f(x) = x(2-x)^2 = x(4-4x+x^2) = 4x-4x^2+x^3$

$$F(x) = 2x^2 - \frac{4}{3}x^3 + \frac{x^4}{4} + C$$

(d) $f(x) = \frac{3}{x} = 3x^{-1}$

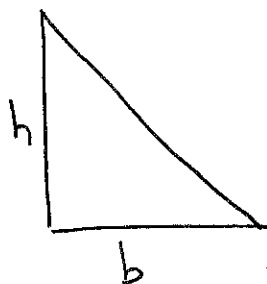
$$\frac{d}{dx} \ln x = \frac{1}{x}$$

$$F(x) = 3 \ln x + C$$

(e) $f(x) = \sqrt[6]{x} - \sec^2(3x) = x^{1/6} - \sec^2(3x)$

$$F(x) = \frac{6}{7}x^{7/6} - \frac{\tan 3x}{3} + C$$

7. {10 points} Recall, the area of a triangle is $A = (\text{base} \cdot \text{height})/2$. A right triangle is enlarged so that its base is growing at a rate of 2 cm/s and its height is growing at a rate of 4 cm/s. How fast is the area growing when the base is 5 cm and the height is 10 cm? {Be sure to show your work.}



$$A = \frac{1}{2} b \cdot h$$

$$\frac{dA}{dt} = \frac{1}{2} h \frac{db}{dt} + \frac{1}{2} b \frac{dh}{dt}$$

$$\frac{db}{dt} = 2$$

$$\frac{dh}{dt} = 4$$

$$= \frac{1}{2} [(10)(2)] + \frac{1}{2} [(5)(4)]$$

$$= 10 + 10 = 20$$

The area is growing at a rate of 20 cm/s when the base is 5 & height is 10.

*{4 points} * Bonus Question: This is for extra credit

Suppose $f'''(x) = \sin(x) + 2$, and $f'''(0) = 1$, $f'(0) = 3$ and $f(0) = -5$. What is $f(x)$?

$$f''(x) = -\cos x + 2x + C = -\cos x + 2x + 2$$

$$f''(0) = 1 = -\cos 0 + 2(0) + C = -1 + C \text{ so } C = 2$$

$$f'(x) = -\sin x + x^2 + 2x + d = -\sin x + x^2 + 2x + 3$$

$$f'(0) = 3 = -\sin 0 + 0^2 + 2 \cdot 0 + d \text{ so } d = 3$$

$$f(x) = \cos x + \frac{x^3}{3} + x^2 + 3x + e$$

$$f(0) = -5 = \cos 0 + 0 + 0 + 0 + e$$

$$\text{so } e = -6$$

$$f(x) = \cos x + \frac{x^3}{3} + x^2 + 3x - 6$$

check. if

$$f(x) = \cos x + \frac{x^3}{3} + x^2 + 3x - 6$$

$$f'(x) = -\sin x + x^2 + 2x + 3$$

$$f''(x) = -\cos x + 2x + 2$$

$$f'''(x) = \sin x + 2 \checkmark$$

10 9

TRIGONOMETRY

Angle Measurement

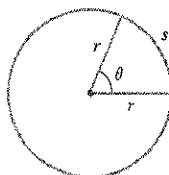
$$\pi \text{ radians} = 180^\circ$$

$$1^\circ = \frac{\pi}{180} \text{ rad}$$

$$1 \text{ rad} = \frac{180^\circ}{\pi}$$

$$s = r\theta$$

(θ in radians)



Right Angle Trigonometry

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

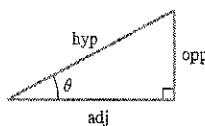
$$\csc \theta = \frac{\text{hyp}}{\text{opp}}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}}$$

$$\sec \theta = \frac{\text{hyp}}{\text{adj}}$$

$$\tan \theta = \frac{\text{opp}}{\text{adj}}$$

$$\cot \theta = \frac{\text{adj}}{\text{opp}}$$



Trigonometric Functions

$$\sin \theta = \frac{y}{r}$$

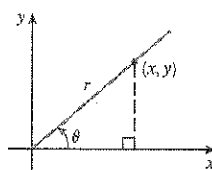
$$\csc \theta = \frac{r}{y}$$

$$\cos \theta = \frac{x}{r}$$

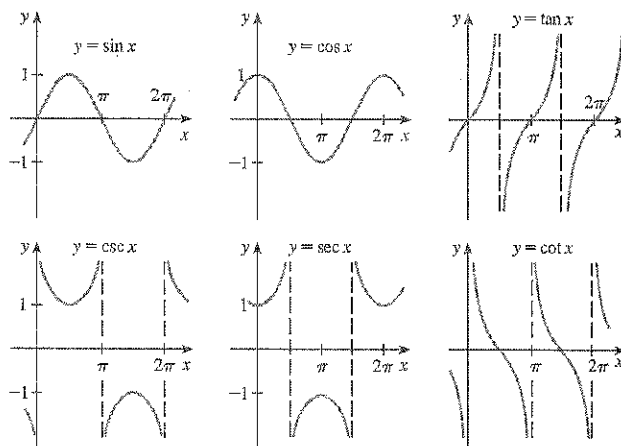
$$\sec \theta = \frac{r}{x}$$

$$\tan \theta = \frac{y}{x}$$

$$\cot \theta = \frac{x}{y}$$



Graphs of Trigonometric Functions



Trigonometric Functions of Important Angles

θ	radians	$\sin \theta$	$\cos \theta$	$\tan \theta$
0°	0	0	1	0
30°	$\pi/6$	$1/2$	$\sqrt{3}/2$	$\sqrt{3}/3$
45°	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1
60°	$\pi/3$	$\sqrt{3}/2$	$1/2$	$\sqrt{3}$
90°	$\pi/2$	1	0	—

Fundamental Identities

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\cot \theta = \frac{1}{\tan \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(-\theta) = -\sin \theta$$

$$\cos(-\theta) = \cos \theta$$

$$\tan(-\theta) = -\tan \theta$$

$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos \theta$$

$$\cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta$$

$$\tan\left(\frac{\pi}{2} - \theta\right) = \cot \theta$$

The Law of Sines

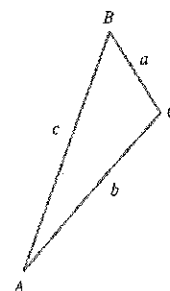
$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

The Law of Cosines

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$



Addition and Subtraction Formulas

$$\sin(x + y) = \sin x \cos y + \cos x \sin y$$

$$\sin(x - y) = \sin x \cos y - \cos x \sin y$$

$$\cos(x + y) = \cos x \cos y - \sin x \sin y$$

$$\cos(x - y) = \cos x \cos y + \sin x \sin y$$

$$\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

$$\tan(x - y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

Double-Angle Formulas

$$\sin 2x = 2 \sin x \cos x$$

$$\cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$$

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$

Half-Angle Formulas

$$\sin^2 x = \frac{1 - \cos 2x}{2} \quad \cos^2 x = \frac{1 + \cos 2x}{2}$$

